The Lefschetz $(1,1)$ Theorem as the First Case of the Hodge Conjecture

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Abstract

The Hodge conjecture predicts that for a smooth projective complex variety, every rational Hodge class of type (p,p) is algebraic. The only fully proven case remains $p = 1$, known as the Lefschetz $(1,1)$ theorem. This article presents a complete, self-contained proof of that theorem for compact Kähler manifolds. The proof combines the exponential sheaf sequence with the Hodge decomposition: the connecting map of the exponential sequence sends holomorphic line bundles to their first Chern class, and the Hodge decomposition shows that every integral $(1,1)$-class lies in the image of this map. We conclude with a discussion of why this argument cannot be generalized to higher p, highlighting the fundamental obstructions discovered by Griffiths.

1 Introduction

Let X be a compact Kähler manifold. The Hodge decomposition expresses the complex cohomology $H^k(X,\mathbb{C})$ as a direct sum

$$H^k(X,\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X),$$

where $H^{p,q}(X)$ is the Dolbeault cohomology of (p,q)-forms. A class $\alpha \in H^{2p}(X,\mathbb{Q})$ is called a *Hodge class* if its image in $H^{2p}(X,\mathbb{C})$ lies entirely in the $H^{p,p}(X)$ component. Algebraic cycles of codimension p (formal linear combinations of subvarieties) give rise to such classes via their fundamental classes, and the *Hodge conjecture* asserts that for a smooth projective variety, every rational Hodge class is a rational linear combination of classes of algebraic cycles.

For $p = 1$, the conjecture is a classical theorem of Lefschetz, usually stated as the $(1,1)$ theorem. It can be phrased in terms of holomorphic line bundles: the first Chern class map

$$c_1 : \mathrm{Pic}(X) = H^1(X, \mathcal{O}_X^*) \longrightarrow H^2(X,\mathbb{Z})$$

has image exactly $H^2(X,\mathbb{Z}) \cap H^{1,1}(X)$, and every such class is the first Chern class of a holomorphic line bundle. Since line bundles correspond to divisors (codimension-1 algebraic cycles), the algebraicity follows.

This note gives a detailed, self-contained proof of the Lefschetz $(1,1)$ theorem for compact Kähler manifolds. The argument proceeds in two natural steps:

1. Use the exponential sheaf sequence to identify the first Chern class with the connecting homomorphism $\delta : H^1(X, \mathcal{O}_X^*) \to H^2(X, \mathbb{Z})$, and verify that the image of this map lies in $H^{1,1}(X)$.

2. Show that the long exact cohomology sequence, together with the Hodge decomposition, forces every integral $(1,1)$-class to be hit by δ, proving surjectivity.

Both steps are elementary and rely only on basic sheaf cohomology and Hodge theory. After the proof, we discuss the obstacles that prevent a similar argument for $p > 1$, thereby illuminating the depth of the general Hodge conjecture.

2 Exponential Sequence and the First Chern Class

On any complex manifold X, there is a short exact sequence of sheaves of abelian groups, the *exponential sequence*,

$$0 \longrightarrow \mathbb{Z}_X \longrightarrow \mathcal{O}_X \xrightarrow{\exp} \mathcal{O}_X^* \longrightarrow 0,$$

where $\mathbb{Z}_X$ is the constant sheaf, $\mathcal{O}_X$ the sheaf of holomorphic functions, $\mathcal{O}_X^*$ the sheaf of nowhere-zero holomorphic functions, and the map $\exp(f) = e^{2\pi i f}$. This sequence is exact because locally every invertible holomorphic function admits a holomorphic logarithm.

Lemma 2.1 (Chern class from the exponential sequence)**.** *For a compact Kähler manifold X, the connecting homomorphism*

$$\delta : H^1(X, \mathcal{O}_X^*) \longrightarrow H^2(X, \mathbb{Z})$$

associated to the exponential sequence sends the isomorphism class of a holomorphic line bundle L to its first Chern class $c_1(L) \in H^2(X, \mathbb{Z})$.

Proof. A holomorphic line bundle L is given by a Čech 1-cocycle $\{g_{\alpha\beta}\}$ with values in $\mathcal{O}_X^*$. Choose a refinement of the cover so that each $g_{\alpha\beta}$ admits a holomorphic logarithm, i.e., we can write $g_{\alpha\beta} = e^{2\pi i f_{\alpha\beta}}$ with $f_{\alpha\beta} \in \mathcal{O}_X(U_\alpha \cap U_\beta)$. The Čech representative of the connecting map δ is obtained by taking the difference of logarithms on triple intersections:

$$n_{\alpha\beta\gamma} = f_{\beta\gamma} - f_{\alpha\gamma} + f_{\alpha\beta}.$$

Then $\{n_{\alpha\beta\gamma}\}$ is a $\mathbb{Z}$-valued 2-cocycle (it is integer-valued because $e^{2\pi i n_{\alpha\beta\gamma}} = 1$), and its cohomology class $\delta(L) \in H^2(X, \mathbb{Z})$ is independent of choices.

Now choose a smooth Hermitian metric on L. Its curvature form Θ is a closed real $(1,1)$-form whose de Rham cohomology class is $c_1(L)_{\mathrm{dR}}$. On the other hand, the Čech-de Rham isomorphism identifies $\delta(L)$ with the same class: the transition functions $g_{\alpha\beta}$ define a connection on L with curvature given by $\bar{\partial}\partial \log |g_{\alpha\beta}|^2$ up to exact forms, and the explicit formula shows that the curvature represents $c_1(L)$. Hence $\delta(L) = c_1(L)$ in $H^2(X, \mathbb{Z})$. A detailed proof can be found in [**?**]. □

Lemma 2.2 (Hodge type of c_1)**.** *For any holomorphic line bundle L on a compact Kähler manifold X, the first Chern class $c_1(L)$ is of Hodge type $(1,1)$, i.e., its image in $H^2(X, \mathbb{C})$ lies in the subspace $H^{1,1}(X)$.*

Proof. By Lemma **??**, $c_1(L)$ is represented by the curvature form Θ of a Hermitian metric on L. This Θ is a real, closed $(1,1)$-form. Under the de Rham isomorphism $H^2_{\mathrm{dR}}(X,\mathbb{C}) \cong H^2(X,\mathbb{C})$, the Dolbeault decomposition of Θ is purely of type $(1,1)$ because a $(1,1)$-form has no $(2,0)$ or $(0,2)$ component. Hence its class lies in $H^{1,1}(X)$. $\square$

Combining the two lemmas, we obtain the well-known factorization:

$$c_1 : \mathrm{Pic}(X) = H^1(X, \mathcal{O}_X^*) \longrightarrow H^2(X,\mathbb{Z}) \cap H^{1,1}(X).$$

Because the right-hand side is a subgroup of $H^2(X,\mathbb{Z})$, the image of c_1 is automatically contained in the intersection.

3 Surjectivity via Hodge Decomposition

The long exact sequence of cohomology beginning from the exponential sequence reads

$$\cdots \longrightarrow H^1(X,\mathcal{O}_X) \longrightarrow H^1(X,\mathcal{O}_X^*) \xrightarrow{\delta} H^2(X,\mathbb{Z}) \longrightarrow H^2(X,\mathcal{O}_X) \longrightarrow \cdots.$$

Let $\alpha \in H^2(X,\mathbb{Z}) \cap H^{1,1}(X)$. We must show that α lies in the image of δ, i.e., that it maps to zero in $H^2(X,\mathcal{O}_X)$. The key is to interpret the map $H^2(X,\mathbb{Z}) \to H^2(X,\mathcal{O}_X)$ through the Hodge decomposition.

Lemma 3.1 (Comparison with Dolbeault cohomology)**.** *For a compact Kähler manifold X, there is a natural isomorphism*

$$H^2(X,\mathcal{O}_X) \cong H^{0,2}(X),$$

where $H^{0,2}(X)$ denotes the Dolbeault cohomology of $(0,2)$-forms. Under this identification and the de Rham isomorphism $H^2(X,\mathbb{C}) \cong \bigoplus_{p+q=2} H^{p,q}(X)$, the map $H^2(X,\mathbb{Z}) \to H^2(X,\mathcal{O}_X)$ induced by the inclusion $\mathbb{Z} \hookrightarrow \mathcal{O}_X$ factors as the composition

$$H^2(X,\mathbb{Z}) \longrightarrow H^2(X,\mathbb{C}) \twoheadrightarrow H^{0,2}(X),$$

where the second arrow is the projection onto the $(0,2)$ Hodge component.

Proof. The Dolbeault resolution $\mathcal{O}_X \to \mathcal{A}^{0,0} \xrightarrow{\bar\partial} \mathcal{A}^{0,1} \xrightarrow{\bar\partial} \cdots$ gives an isomorphism $H^q(X,\mathcal{O}_X) \cong H^{0,q}_{\bar\partial}(X)$. For $q=2$, this is $H^{0,2}(X)$. The map $H^2(X,\mathbb{Z}) \to H^2(X,\mathcal{O}_X)$ comes from the inclusion of the constant sheaf $\mathbb{Z}$ into $\mathcal{O}_X$. In the Dolbeault picture, the inclusion $\mathbb{Z} \hookrightarrow \mathbb{C} \hookrightarrow \mathcal{O}_X$ factors through the sheaf of locally constant functions, and the induced map on cohomology sends an integral class to its de Rham class (in $H^2(X,\mathbb{C})$) followed by the projection onto the $H^{0,2}$ component, because the Dolbeault cohomology $H^{0,2}$ is canonically a direct summand of $H^2(X,\mathbb{C})$. More precisely, the composition

$$H^2(X,\mathbb{Z}) \to H^2(X,\mathbb{C}) \to H^2(X,\mathcal{O}_X) \cong H^{0,2}(X)$$

is exactly the projection, since the Hodge filtration satisfies $F^2H^2 = H^{2,0} \oplus H^{1,1}$ and $H^2(X,\mathcal{O}_X) \cong H^2/F^2$ is isomorphic to $H^{0,2}$. See [**?**]. $\square$

Lemma 3.2 (Vanishing in $H^{0,2}$)**.** *Let $\alpha \in H^2(X,\mathbb{Z}) \cap H^{1,1}(X)$. Then the image of α in $H^2(X,\mathcal{O}_X)$ is zero.*

Proof. By Lemma **??**, the image of α in $H^2(X,\mathcal{O}_X)$ is identified with its $(0,2)$ Hodge component. By assumption, α is of pure type $(1,1)$, so its $(0,2)$ component is zero. Hence it maps to 0. □

We can now prove the main surjectivity statement.

Proposition 3.3 (Lifting integral $(1,1)$-classes)**.** *Every class $\alpha \in H^2(X,\mathbb{Z}) \cap H^{1,1}(X)$ is the first Chern class of some holomorphic line bundle on X.*

Proof. Consider the long exact sequence

$$\cdots \to H^1(X,\mathcal{O}_X^*) \xrightarrow{\delta} H^2(X,\mathbb{Z}) \xrightarrow{\rho} H^2(X,\mathcal{O}_X) \to \cdots .$$

By Lemma **??**, $\rho(\alpha) = 0$. Therefore α lies in the kernel of ρ, which by exactness equals the image of δ. Hence there exists $L \in H^1(X,\mathcal{O}_X^*)$ such that $\delta(L) = \alpha$. By Lemma **??**, $\delta(L) = c_1(L)$, so $\alpha = c_1(L)$. □

4 Main Result: The Lefschetz $(1,1)$ Theorem

We now assemble the pieces into the classical theorem.

Theorem 4.1 (Lefschetz $(1,1)$ theorem)**.** *Let X be a compact Kähler manifold. Then the first Chern class map*

$$c_1 : \mathrm{Pic}(X) \longrightarrow H^2(X,\mathbb{Z}) \cap H^{1,1}(X)$$

is surjective. Equivalently, every integral $(1,1)$-class on X is the first Chern class of a holomorphic line bundle, hence is algebraic.

Proof. Lemma **??** and Lemma **??** show that c_1 lands in the claimed subgroup. Proposition **??** proves that the map is surjective onto that subgroup. □

Remark 4.2. For a smooth projective variety X, the Picard group $\mathrm{Pic}(X)$ is isomorphic to the group of divisors modulo linear equivalence, and the first Chern class of a divisor is its fundamental class. Therefore the theorem asserts that every integral $(1,1)$-class arises from a divisor, confirming the Hodge conjecture for $p = 1$.

5 Discussion and Open Problems

The above proof uses two ingredients that are specific to $p = 1$ and fail for higher p:

1. **The exponential sequence.** For line bundles, there is the short exact sequence $0 \to \mathbb{Z} \to \mathcal{O}_X \to \mathcal{O}_X^* \to 0$ whose cohomology links $H^1(X,\mathcal{O}_X^*)$ (the Picard group) to $H^2(X,\mathbb{Z})$ and $H^2(X,\mathcal{O}_X)$. No analogous sequence is known for higher codimension. Codimension p cycles are parametrized by Chow groups $\mathrm{CH}^p(X)$, whose relation to sheaf cohomology is far more complicated. In particular, there is no sheaf whose H^{2p} receives the cycle class map and whose higher direct images can be used to lift classes in a way parallel to the $p = 1$ case.

2. **The Hodge decomposition at level 2.** The map $H^2(X, \mathbb{Z}) \to H^2(X, \mathcal{O}_X)$ factors through projection onto $H^{0,2}(X)$. For a $(1,1)$-class, this projection vanishes. For $p > 1$, any rational (p,p)-class has trivial projection onto $H^{p+q,p-q}(X)$ for $q \neq 0$, but the analogous map from $H^{2p}(X, \mathbb{Z})$ to some cohomology group of a sheaf would need to capture exactly the non-(p,p) Hodge components. There is no such sheaf; the natural recipient of the cycle class map is the Deligne cohomology or the intermediate Jacobian, which involves all Hodge components and is far from a simple projection.

3. **Obstructions and the general Hodge conjecture.** For $p \geq 2$, not every integral (p,p)-class is algebraic. Griffiths [**?**] constructed classes that are Hodge classes yet not algebraic, showing that the Hodge conjecture must be formulated over $\mathbb{Q}$ (rational coefficients) and that integrality is not sufficient. Moreover, the failure of the naive lifting argument is explained by the theory of variations of Hodge structure and the fact that the Hodge conjecture is equivalent to the statement that the Abel–Jacobi map onto the intermediate Jacobian has certain surjectivity properties, which are known to be false in general. The minimal dimension for a counterexample to an integral Hodge conjecture is 4, while the rational Hodge conjecture remains open for all $p > 1$.

Thus, while the Lefschetz $(1,1)$ theorem is a beautiful and elementary entry point to Hodge theory, the general Hodge conjecture demands a deeper understanding of algebraic cycles and Hodge structures that remains at the frontier of current research.

References

[1] P. Griffiths and J. Harris, *Principles of Algebraic Geometry*, Wiley, 1978.

[2] C. Voisin, *Hodge Theory and Complex Algebraic Geometry I*, Cambridge University Press, 2002.

[3] P. Griffiths, *On the periods of certain rational integrals I, II*, Ann. of Math. **90** (1969), 460–495; **90** (1969), 496–541.